

The paper-correct HCAPO still drifts:

why a faithful implementation of Eq. 6–7 lengthens its own replies

Agentic-RL-CA project

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Abstract

This note is about *one* run: HCAPO implemented exactly as the paper specifies (arXiv:2603.08754, Eq. 6–7 — verified: the credit ratio ρ centred on 1.0, two-sided, clip engaging on both ends). It trained cleanly for ~ 150 steps, then its replies quadrupled in 40 steps and pinned against the 2048-token cap, with no accuracy gain. The drift is not our earlier port bug (that was found and fixed first); it follows from two properties of the formulation itself: (i) a turn’s hindsight score is a *per-token average*, which rises when a turn is padded with easy filler — an incentive we measure in healthy-length rollouts *before any drift exists* (Fig. 3); and (ii) the trajectory-average normalisation makes credit *relative*, which cancels collective lengthening but not the batch-by-batch fact that the *longer sibling wins* — so each update nudges the policy longer, permanently, while the normalisation guarantees nobody ever nets credit from it (Fig. 2). The race only ends when padding has homogenised all turns and the credit signal itself dies (Fig. 4).

1 What happened

We verified the implementation before this run: ρ starts centred on 1.0 (batch mean 1.0006), takes values on both sides of 1 (both clip bounds engage), and a unit test confirms that lengthening *all* turns together leaves ρ exactly unchanged — the invariance the paper’s intra-trajectory normalisation is designed to provide. The drift happened anyway.

2 Why: the incentive exists before the drift

HCAPO scores a turn by its *average* per-token log-probability under a hindsight-augmented prompt (Eq. 6), then credits each turn by its score *relative to the trajectory’s average* (Eq. 7). Averages can be padded: the tokens that carry a turn’s content are hard (low probability), filler is easy (high probability), so adding filler raises the average. The question is whether this bites *in practice*, in the healthy regime, within Eq. 7’s normalisation. We measured it directly — 145 healthy-length trajectories, every turn re-scored with and without the hindsight text, everything compared *within* its own trajectory:

Two details make this worse than a generic length bias. First, the plain (no-hindsight) score shows *no* such correlation ($r = -0.02$) — the length preference is created by the hindsight injection itself: appending the episode’s final observation perturbs early tokens most, and longer turns dilute that perturbation away. Second, it survives position controls, so it is not an artifact of later turns being longer.

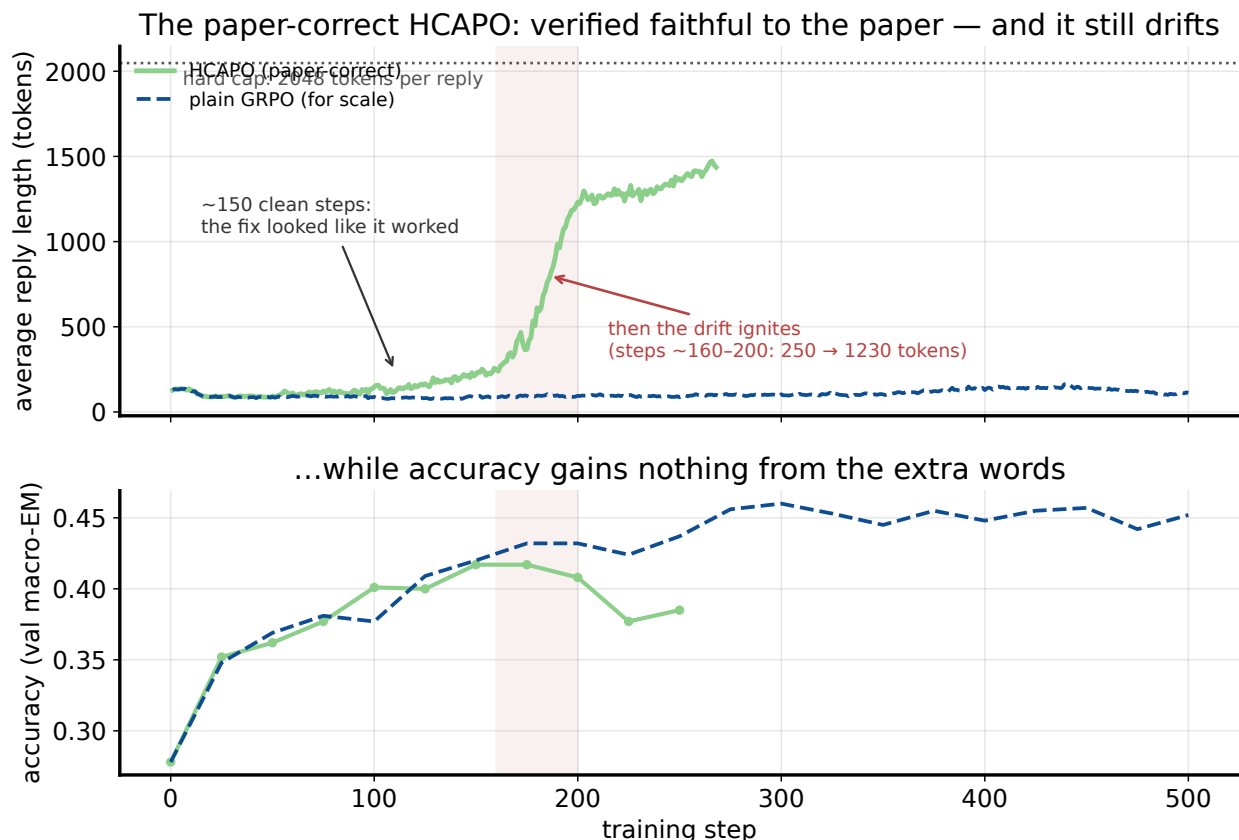


Figure 1: Top: average reply length. The paper-correct run behaves normally for ~ 150 steps — indistinguishable from GRPO — then drifts from ~ 250 to $\sim 1,230$ tokens between steps 160 and 200 and saturates against the cap. Bottom: accuracy over the same steps — flat through the entire drift, and never above plain GRPO.

3 Why the normalisation doesn't save you

The invariance argument (“if everyone lengthens together, ρ is unchanged”) is true and we verified it — but training never moves in lockstep. Gradients act on *individual sampled turns*, and Fig. 3 shows which individual turns win: the longer ones. The uniform direction being invariant means precisely that there is *no restoring force*: collective lengthening is free. Relative credit plus an absolute lever (padding raises your own score) is an arms race.

4 How it ends: the signal eats itself

The endpoint is self-limiting rather than catastrophic: accuracy holds around 0.38–0.42 while more than half of all turns are cut off mid-sentence by the cap, and each training step takes $\sim 2.6\times$ longer than before the drift.

What would actually fix it

(1) Score turns on a *fixed window* or another length-robust statistic, so filler cannot move the score — this removes the lever the ratchet needs. (2) Exclude the final (answer) turn from hindsight



Figure 2: Each point is one turn, positioned by how much longer it is than its trajectory’s average (x) and how much higher its hindsight score-lift is than the trajectory’s average (y). The correlation is $r = +0.43$ (and $+0.33$ after controlling for turn position): **before any drift exists, the relatively longer turn already gets the relatively higher hindsight score.** This is exactly the quantity Eq. 7 turns into credit.

credit; separate measurements show it is the top-credit turn in 73% of healthy-regime trajectories purely because the ending entails it. (3) A per-turn length penalty would mask the symptom but leave the mis-scoring in place.

Fine print

One seed; SearchQA (Qwen3-4B, non-thinking, 4-turn episodes, 2048-token cap). The paper’s own benchmarks (ALFWorld, WebShop) use short action-style turns under tight caps and may never enter this regime; this note is a claim about the formulation on this task class, not a refutation of the paper’s results. The run is still training; step-500 finals will be appended. Companion documents: the full-family runaway report (2026-07-30_hcapo_length_runaway/), the technical implementation note (2026-07-29_hcapo_implementation_note/), and the timestamped prediction/retraction trail (2026-07-21_4b_horizon_eval/results/hcapo_wave3_preregistration.md). Rebuild: `assets/build_figures.py`, then `tectonic report.tex`.

Why the trajectory-average normalisation doesn't save you: credit is relative, but the policy update is permanent

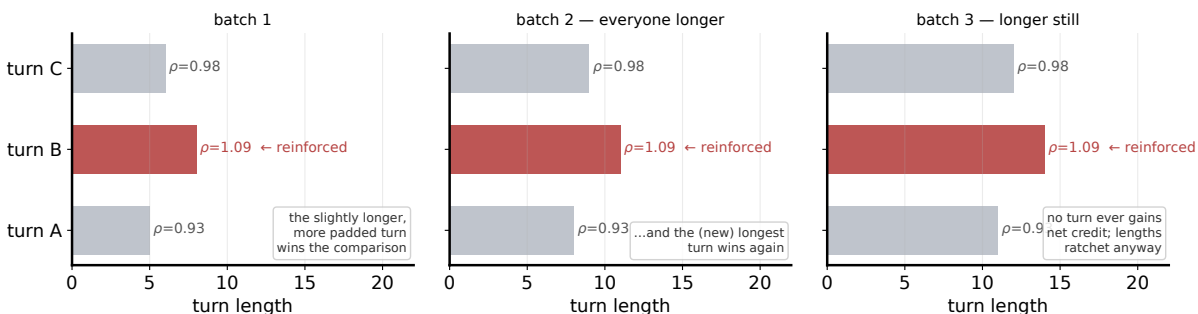
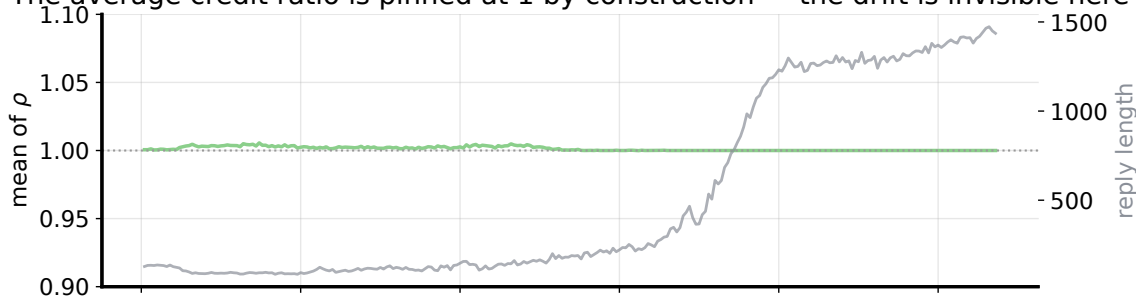


Figure 3: The treadmill. In every batch, the turn that happens to be a little longer and more padded than its trajectory-siblings gets $\rho > 1$ and is reinforced (red). The policy shifts longer *everywhere*; the normalisation re-centres, so nobody keeps any credit advantage — but the behavioural update is permanent, and at the new, longer baseline the same comparison favours the same thing. Grading on a curve does not stop grade inflation; it only guarantees the inflation buys nothing.

The average credit ratio is pinned at 1 by construction — the drift is invisible here



...but the SPREAD — the credit signal's entire information content — collapses as padding homogenises the turns (0.135 $\rightarrow$ 0.030): the race ends when everyone looks the same

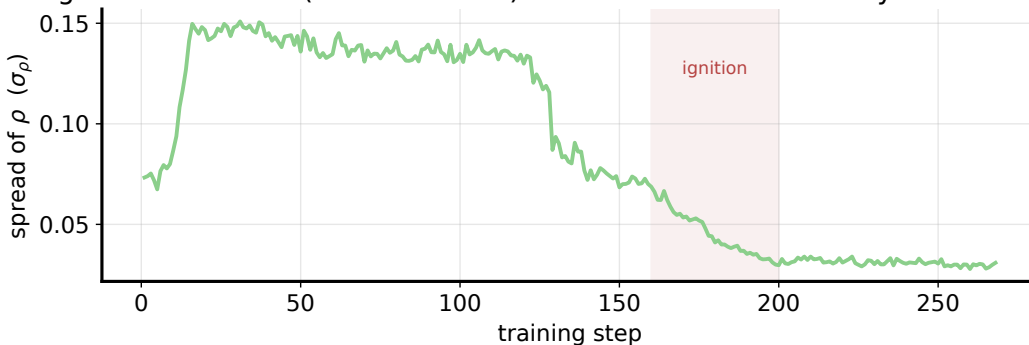


Figure 4: Top: the batch *mean* of ρ is pinned at 1.0 by construction throughout — monitoring the mean can never see this failure. Bottom: the *spread* of ρ — the method's entire information content — collapses from 0.135 to 0.030 exactly across the drift window. As padding homogenises the turns, every turn's average score converges to the same generic fluency level, all credit ratios approach 1, and the hindsight term goes silent. What remains is plain GRPO plus a small position bonus, at $\sim 12\times$ the token cost.